

Robot Dynamics & Control

Assignment 3: Dynamic Robot Control

May, 19th, 2026

Important notes (please read carefully):

- Your Simscape models must have the same configurations as the figures, with the same parameters shown;
- Your Simulink files must be uploaded/exported in Matlab R2025a or lower versions.

3R Planar Robot (3R)

Initial Joint configurations shown in the figure: $q_1=q_2=q_3=0^\circ$.

The robot configuration is shown in the last page.

You must start from the planar 3R robot model shown in the figure, which follows the **Denavit-Hantherberg convention**. The **base** frame is rotated with respect to the **world** frame as indicated in the following expression:

$$R_b^w = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Link Parameters

Length of Link 1: 20 cm

Length of Link 2: 20 cm

Length of Link 3: 20 cm

Width/Height of the Links: 1 cm

Moments of inertia of the Links : You can consider only i_{zz1} , i_{zz2} , i_{zz3} since the others are close to 0 kg·m²

A3.1 Question (PD control)

Mass of Link 1: 2.5 kg

Mass of Link 2: 4.5 kg

Mass of Link 3: 1.5 kg

Mass of end-effector: 0 kg

Mass of all joints: 0 kg

Initial Joint configuration : $q_i = [q_1, q_2, q_3] = [10^\circ, 20^\circ, -30^\circ]$

Proportional gain : 5.0

Derivative gain : 4.5

The robot must be able to reach the following final configuration within **10 seconds**:

Final Joint configuration : $q_f = [q_1, q_2, q_3] = [40^\circ, -10^\circ, -50^\circ]$

Do not consider the **damping effects** for all of three joints.

At **t=10s**, compute the **Desired torque τ (in $N \cdot m$)** of the Joints w.r.t their own frames, to reach the desired configuration.

A3.2 Question (Computed Torque Control)

Mass of Link 1: 2.5 kg

Mass of Link 2: 4.5 kg

Mass of Link 3: 1.5 kg

Mass of end-effector: 0 kg

Mass of all joints: 0 kg

Initial Joint configuration : $q_i = [q_1, q_2, q_3] = [10^\circ, 20^\circ, -30^\circ]$

Proportional gain : 5.0

Derivative gain : 4.5

The robot must be able to reach the following final configuration in **10 seconds**, using a **polynomial trajectory block**:

Final Joint configuration : $q_f = [q_1, q_2, q_3] = [30^\circ, 40^\circ, -10^\circ]$

Do not consider the **damping effects** for all of three joints.

At **t=10s**, compute the **Desired torque τ** (in **N · m**) of the Joints w.r.t their own frames, required to follow the trajectory.

